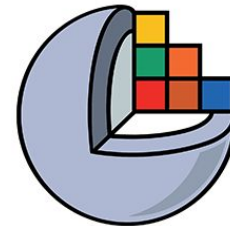
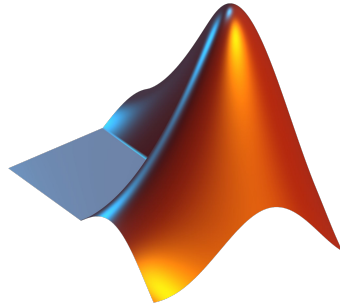


Morphologies Measurement Protocol Notes

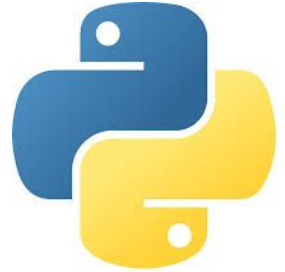
O'Connell Biomechanics Lab

Yousuf + Sylvi

Winter 2025



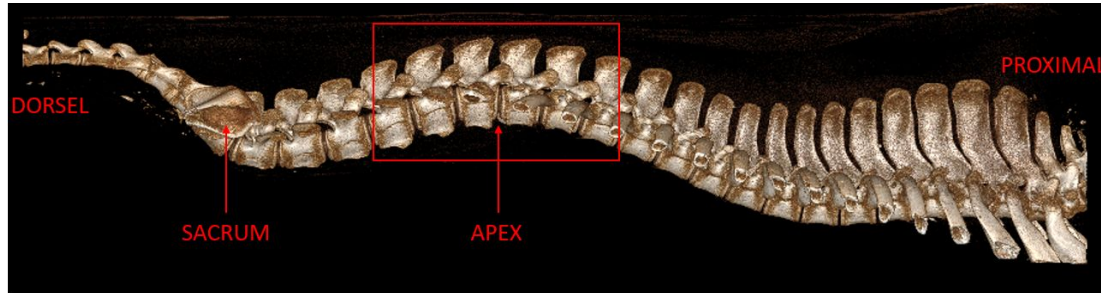
3D Slicer



Berkeley
UNIVERSITY OF CALIFORNIA

Measurement Protocol Overview

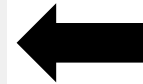
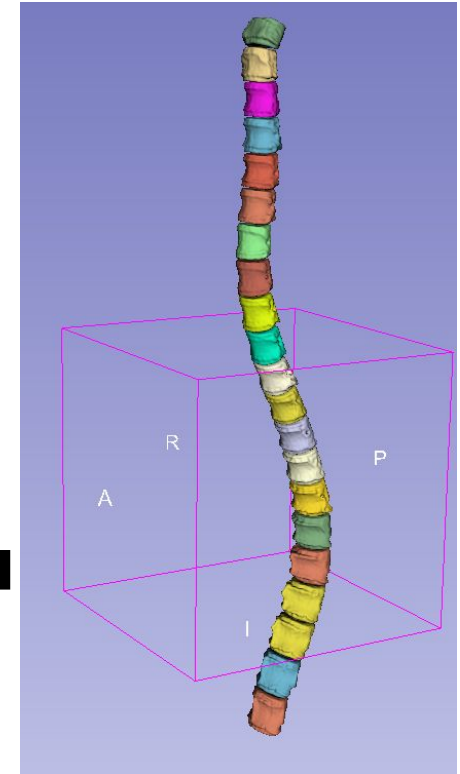
Raw imaging data



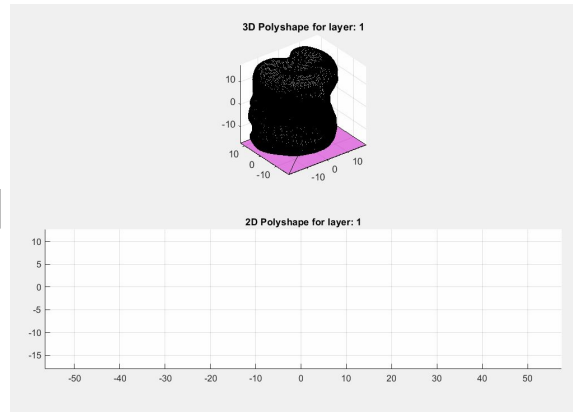
Assumes: 15
thoracic + 6
lumbar levels



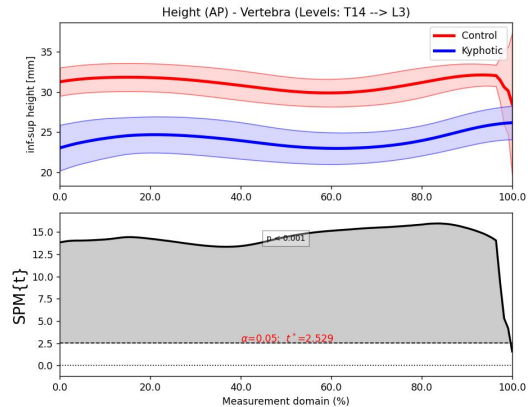
Full spine segmentation^[1]



(Automated) MATLAB measurements



Statistical Analysis



[1] Manual segmentation protocols [here](#)

Programing Overview

Morphologies Github [here](#)^[1-2], general pipeline:

Loading all vertebrae geometry files → disc construction → geometry alignment → slicer, height, and volume measurements → analysis

Morphologies

Author: Yousuf Abubakr (yousufabubakr123@berkeley.edu)

Lab: Grace O'Connell Biomechanics Lab (<https://oconnell.berkeley.edu/>)

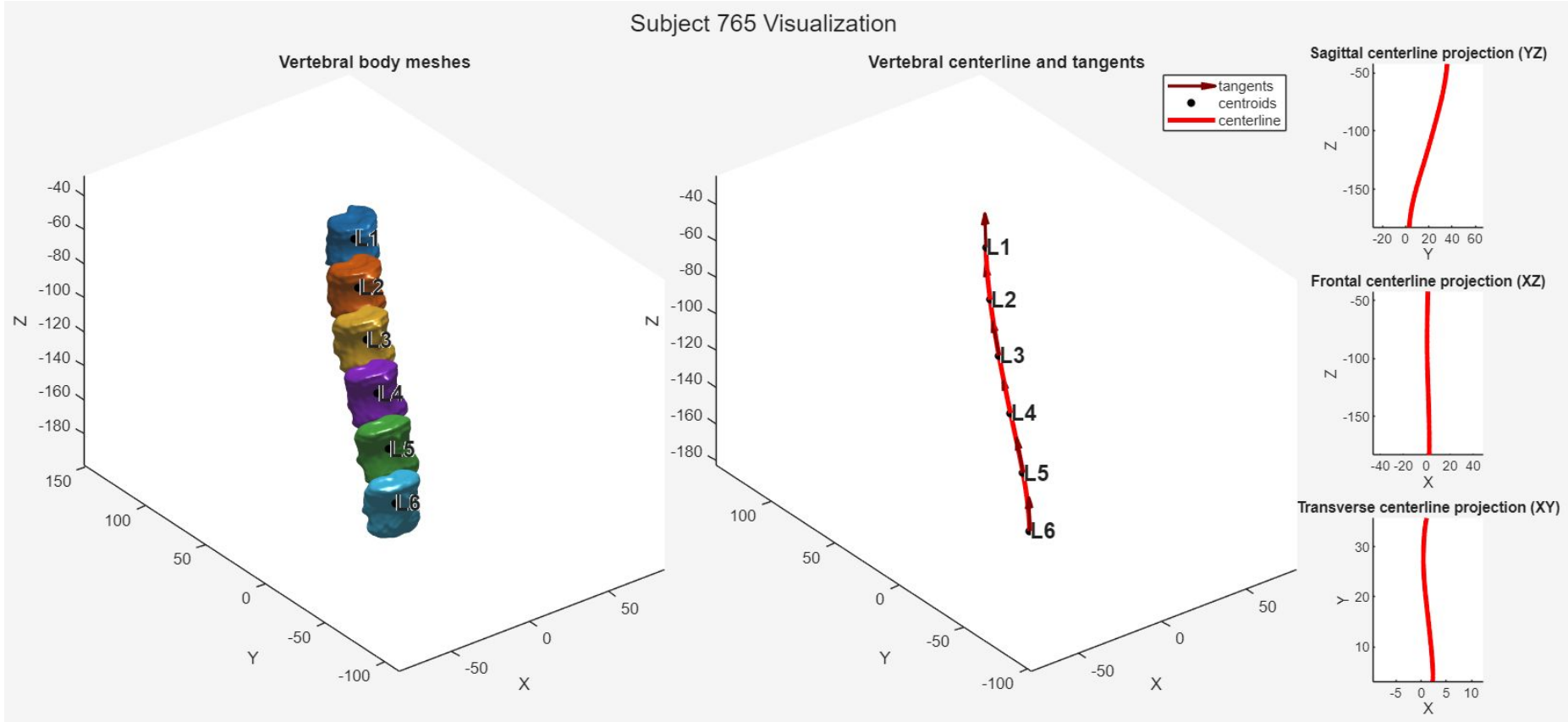
Description: A toolkit for processing, analyzing, and visualizing morphological data from medical imaging datasets (e.g., STL meshes, MATLAB measurement files).

[1] Github stats (as of 6/28/2026): total # lines of code = 6,177, total # of words = 27,111, total # of characters = 240,411, total # of MATLAB .m files = 73

[2] Necessary MATLAB packages: Curve Fitting Toolbox

MATLAB Program Overview

1.) Loading geometry: *loads vertebral body geometries & computes centerline path and tangents*



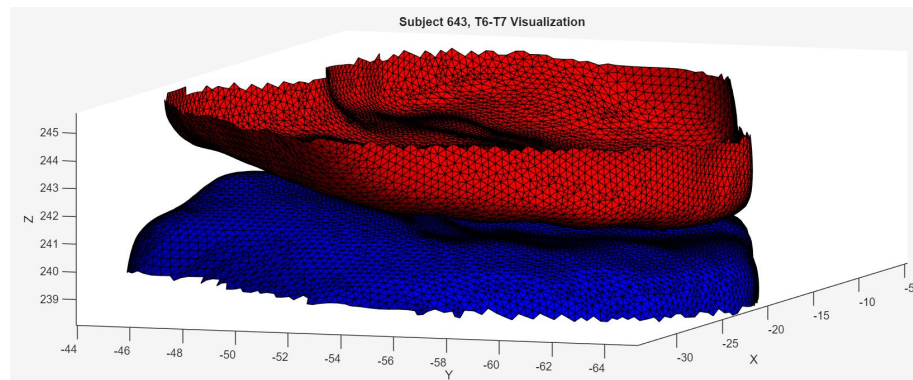
MATLAB Program Overview

2.) Disc construction: *interpolating across vertebral endplates to define and export disc volumes*

Step 1:

Extract triangulation representations of superior + inferior surfaces

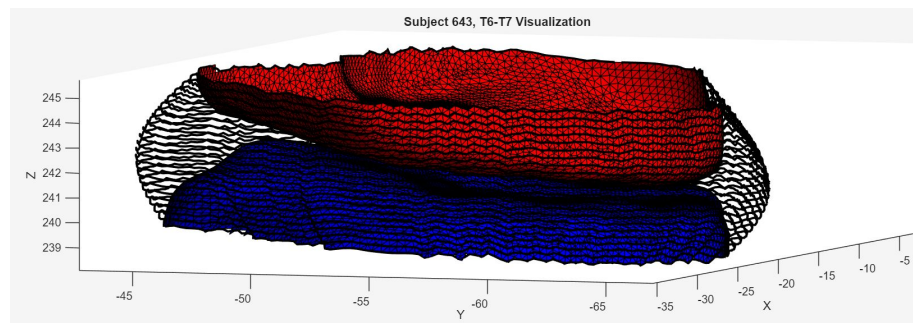
- Red = superior surface of disc
- Blue = inferior surface of disc



Step 2:

Obtain inferior → superior loft curves (pictured in black)

- Associated parameters:
 - # of rings
 - bulge amplitude (default: 2 mm)



MATLAB Program Overview

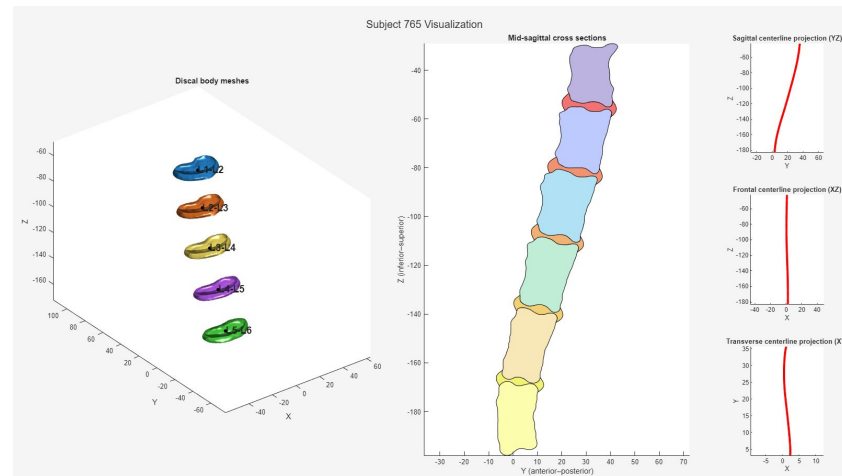
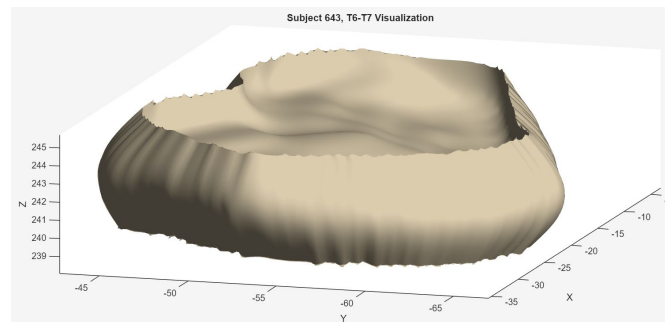
2.) Disc construction: *interpolating across vertebral endplates to define and export disc volumes*

Step 3:

Stitch endplate surfaces to one another to create a full disc triangulation and export to .stl file

NOTES:

- **Water-tightness is guaranteed**
- Any further geometry processing and measurement processes are generalized for both vertebra and disc structures

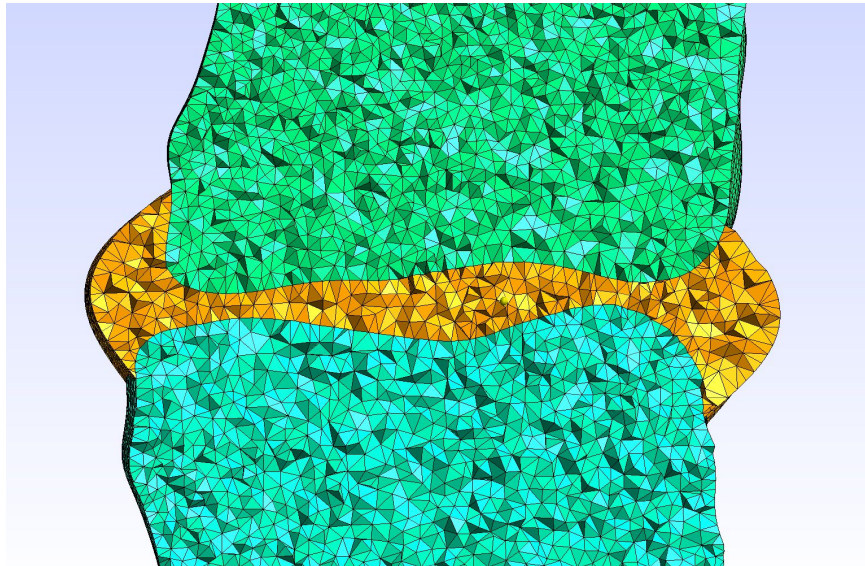


MATLAB Program Overview

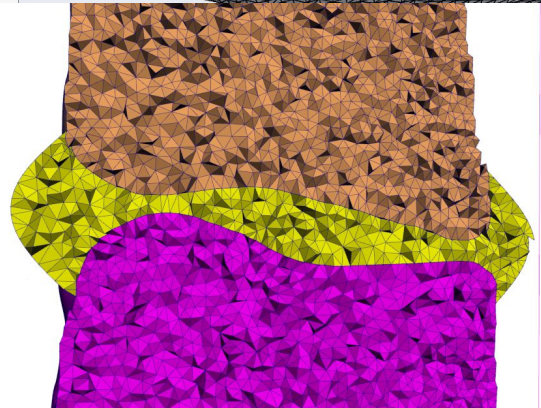
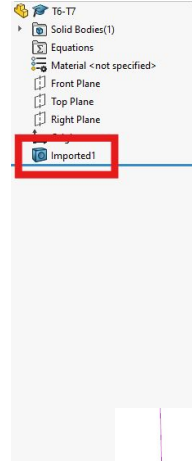
2.) Disc construction: *interpolating across vertebral endplates to define and export disc volumes*

NOTES:

- Assembly meshing in [Gmsh](#) (following a protocol similar to [this](#)) is easy!

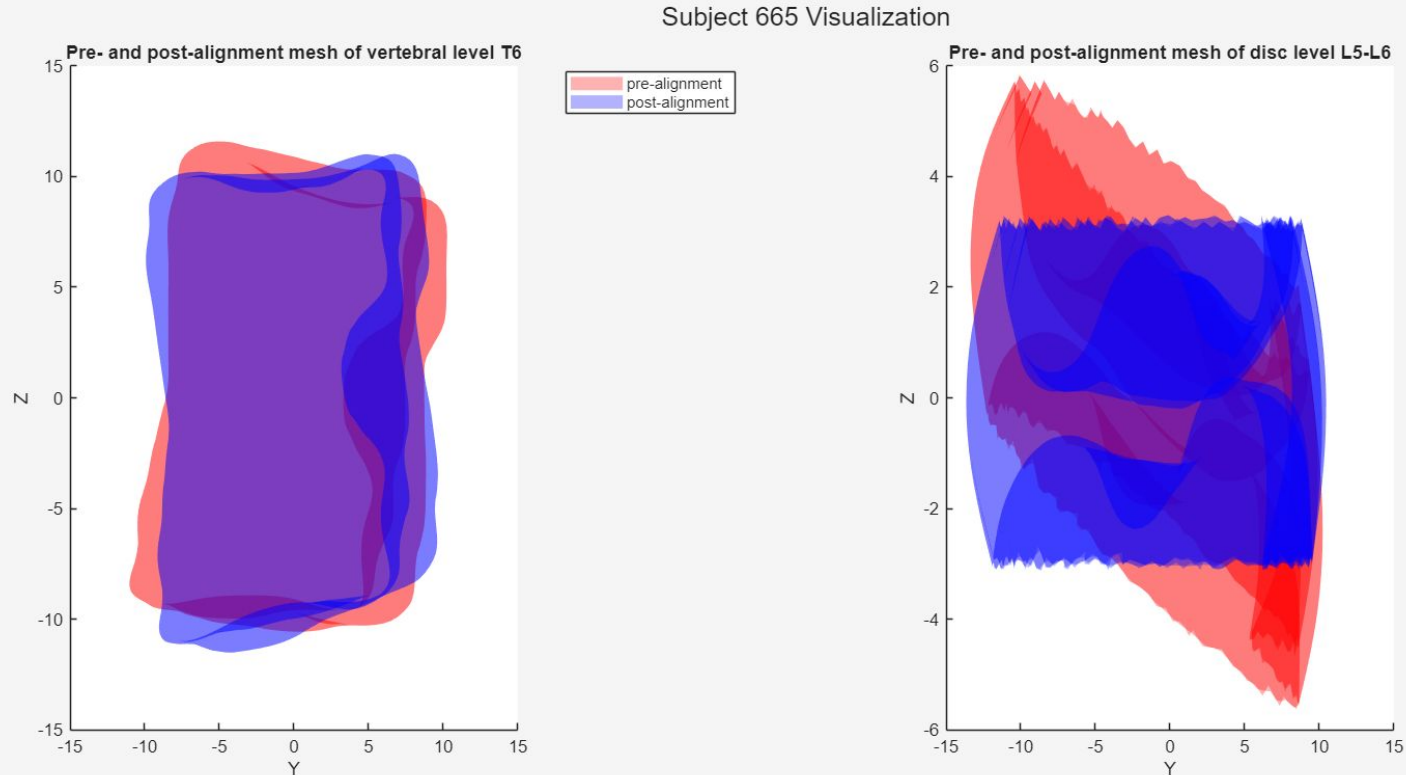


Importing to
FEBio!



MATLAB Program Overview

3.) Geometry alignment: *centerline-based geometry alignment to standard coordinate reference frame*



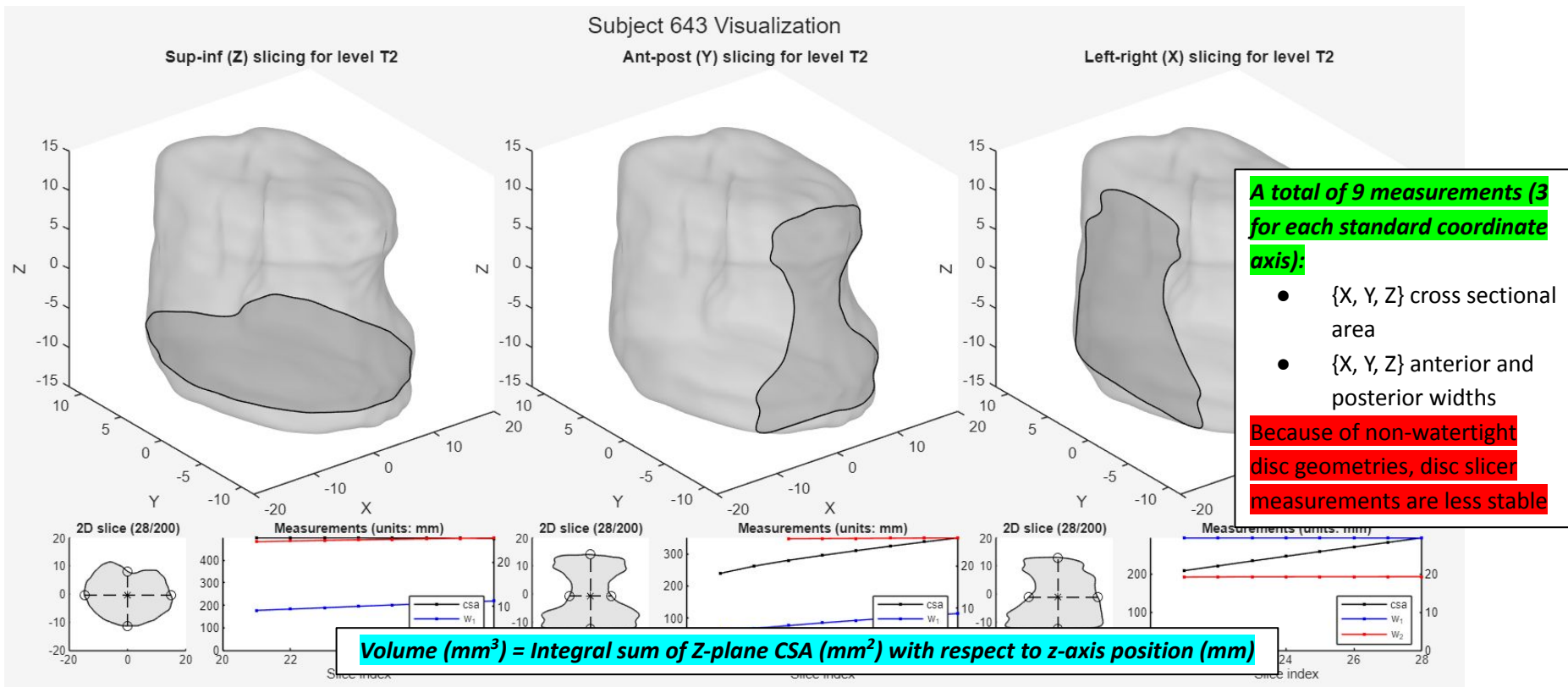
Rotation alignment matrix is found from centerline geometry:

- 3D coordinate frame is determined at every disc/vertebra centroid-centerline position
- 3D coordinate frame $\rightarrow$ basis of rotation matrix
- Every disc/vertebrae geometry is transformed via the formula:

$$\circ \quad V_I = (V_0 + t) * R$$

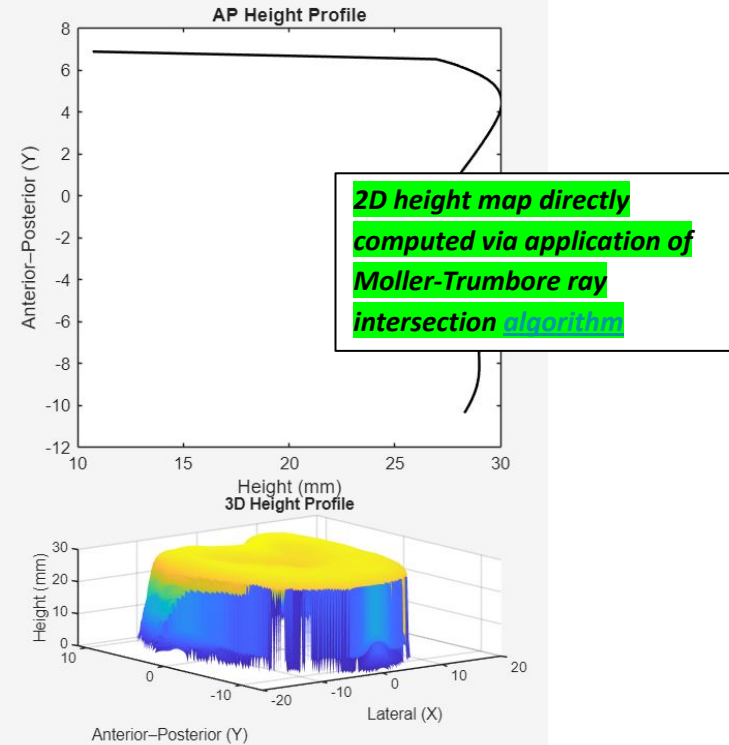
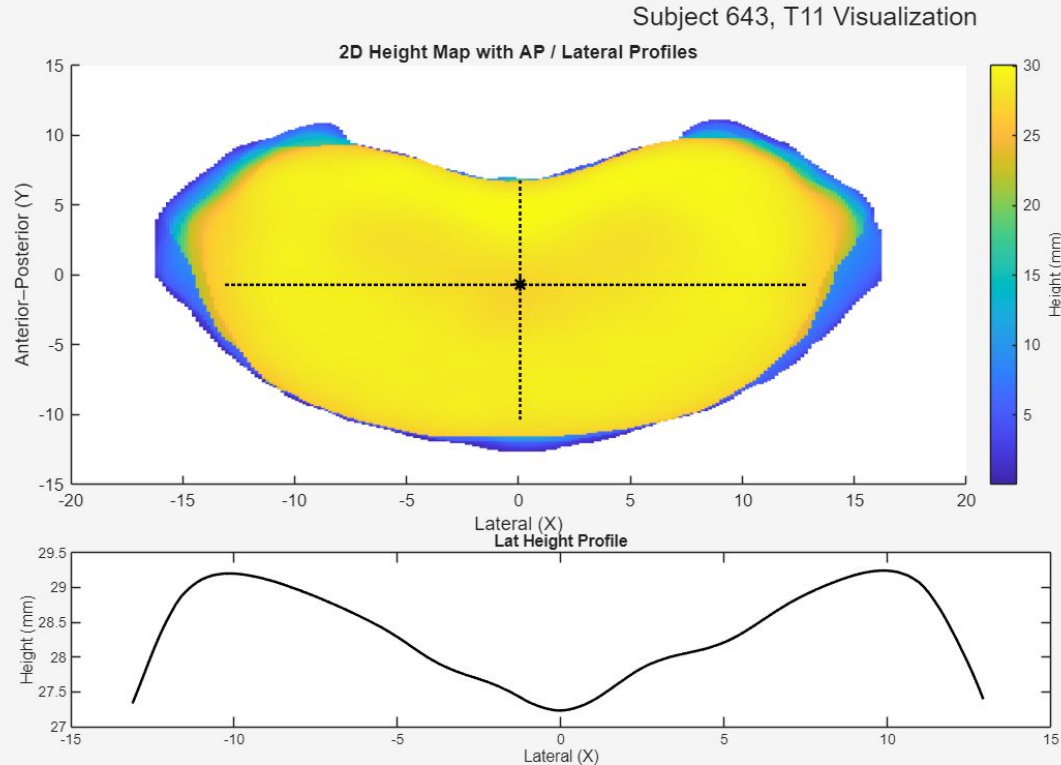
MATLAB Program Overview

4.) Slicer/volume measurements: *slicing geometries and measuring CSAs, widths, and volumes*



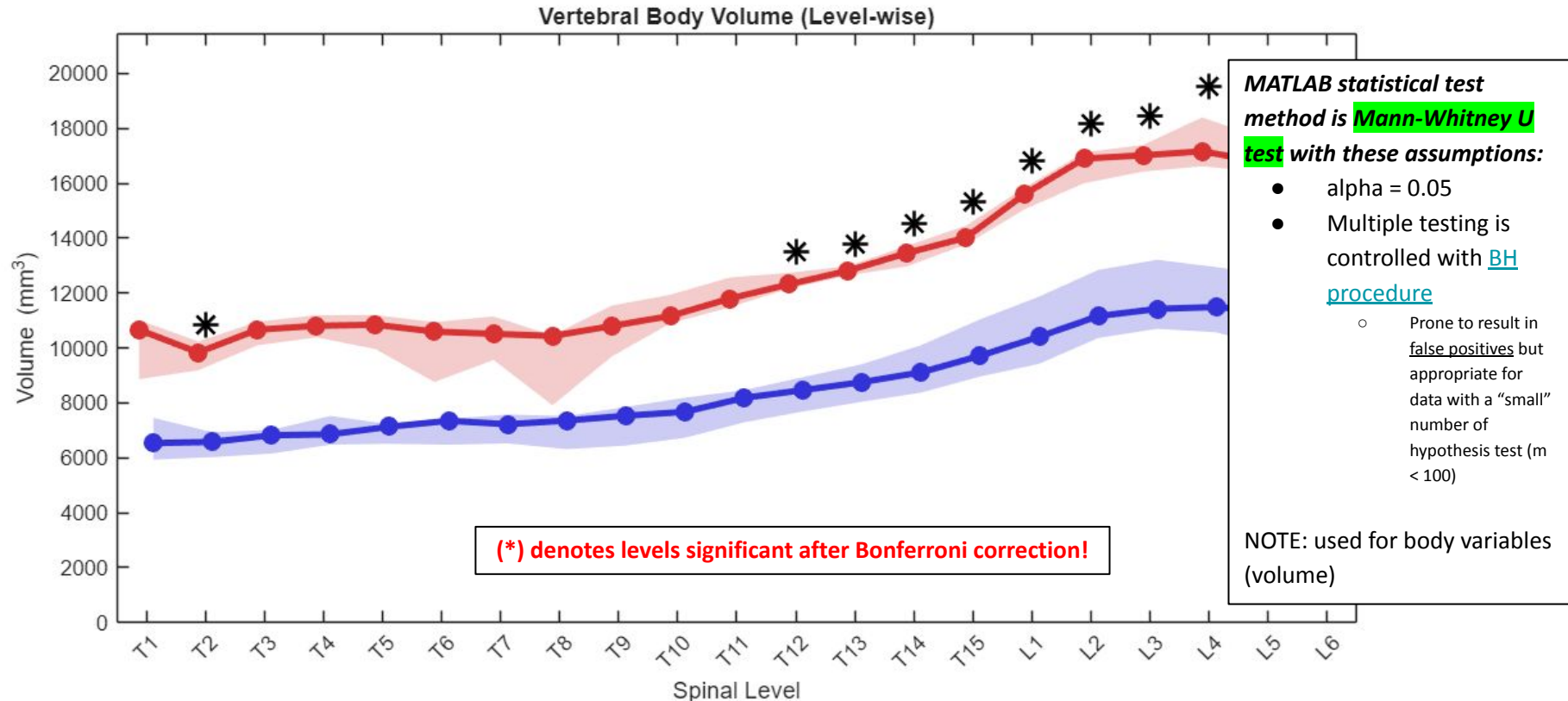
MATLAB Program Overview

4.) Height measurements: *measuring 2D height distribution and extracting AP and LAT heights*



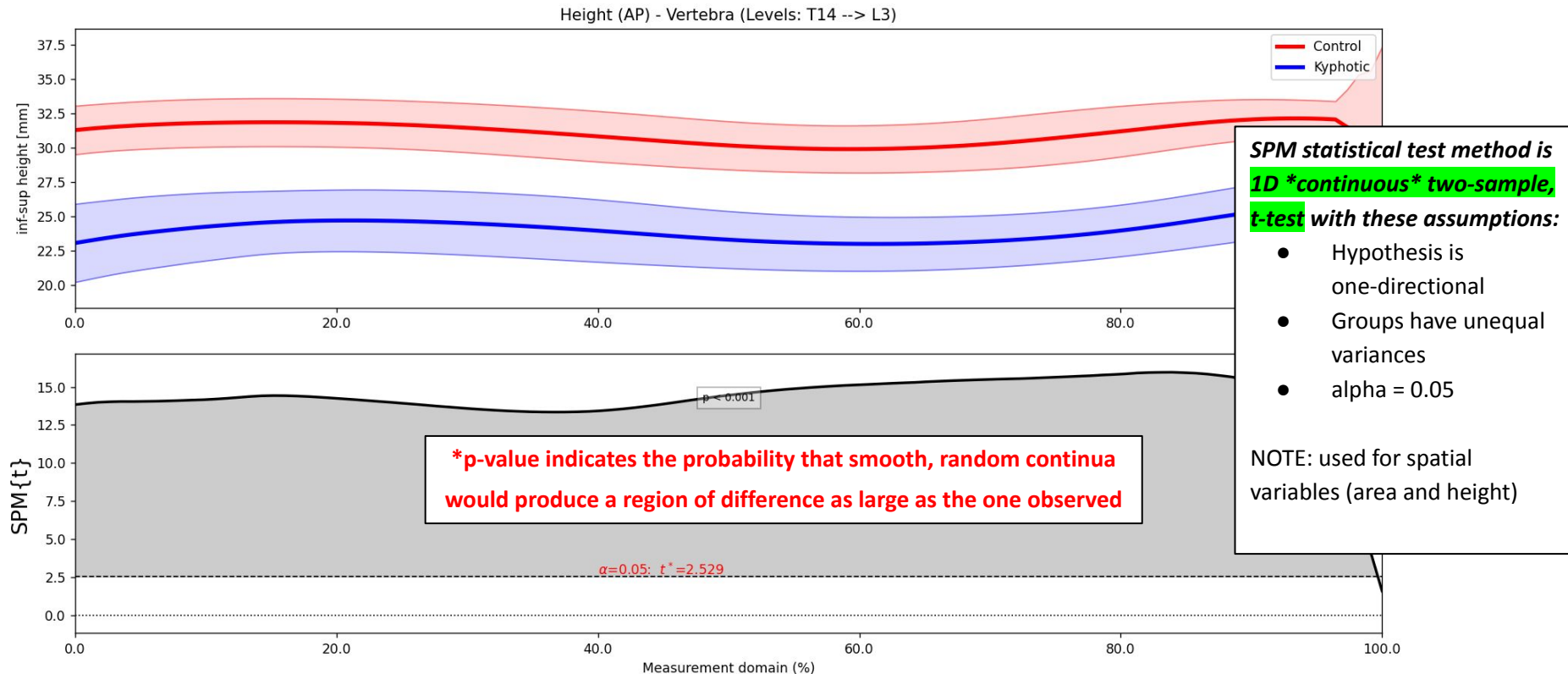
MATLAB Program Overview

5.) Analysis: *computing and visualizing summary statistics across control and kyphotic experimental groups*



MATLAB Program Overview

5.) Analysis: *computing and visualizing summary statistics across control and kyphotic experimental groups*



Morphologies Technical Overview

Morphologies source code [here](#), head files:

main.m	←-----	Execution hub of all programs and subprograms
makeConfig.m	←-----	Configuration file for pipeline (# of slices, height resolution, etc)
Primary routine files (executed in main.m)		setSubjectInformation.m ←----- initializes pipeline data structures
		loadGeometryMetadata.m ←----- loads geometry into data structure
		constructDiscs.m ←----- disc construction
		alignGometries.m ←----- aligns vertebrae and disc geometries
		makeSlicerMeasurements.m ←----- makes slicer measurements
		makeHeightMeasurements.m ←----- makes height measurements
		makeVolumeMeasurements.m ←----- makes volume measurements
		exportData.m ←----- writes data to file
		summarizeData.m ←----- visualizes and exports data

Morphologies Technical Overview

Morphologies source code [here](#), head files:

main.m	←-----	Execution hub of all programs and subprograms
makeConfig.m	←-----	Configuration file for pipeline (# of slices, height resolution, etc)

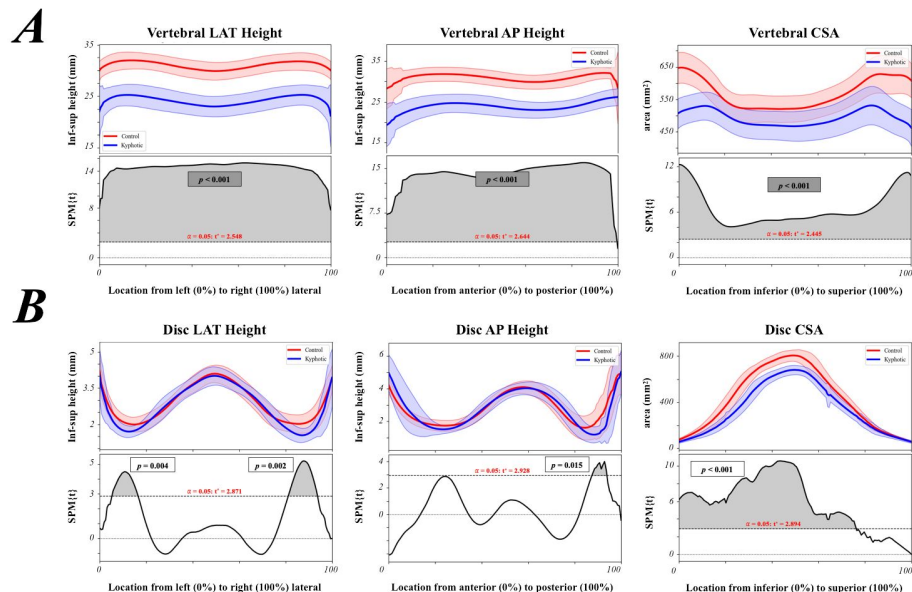
How to use source code:

1. *Set configuration settings in makeConfig.m*
 - a. **IMPORTANT PARAMETERS:**
 - i. **numSlices, heightResolution, slicerIgnorance, heightIgnorance**
 - ii. *If pipeline detects any of these config parameters are different from the saved config settings in 'data/raw', the measurements will be rewritten!*
 - b. *Options for visualization (note: some plots are graphics intensive)*
 - c. *Disc construction parameters*
 - d. *Range of levels to be exported for SPM analysis*
2. *Run main.m for MATLAB results and run '[analysis-utils/main.py](#)' for SPM results*

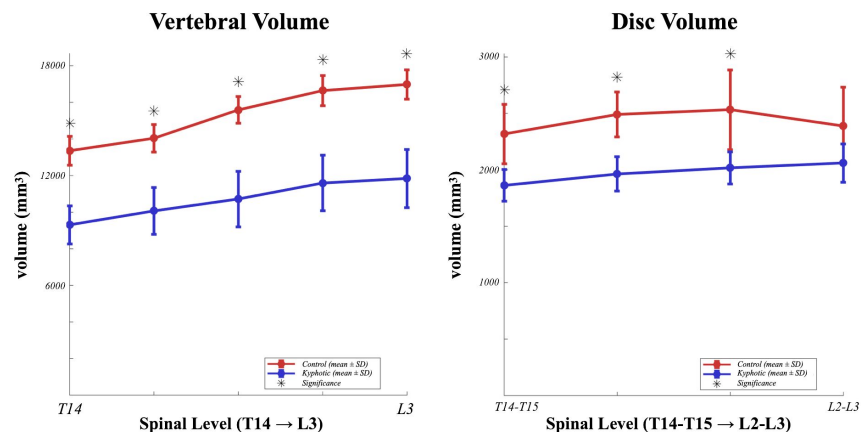
Summary of Results from *Morphologies*

Settings: # of slices = 200, height resolution = 200, slicer & height ignorance = 0.1, levels exported = T14 → L3

Spatially-varying measurements



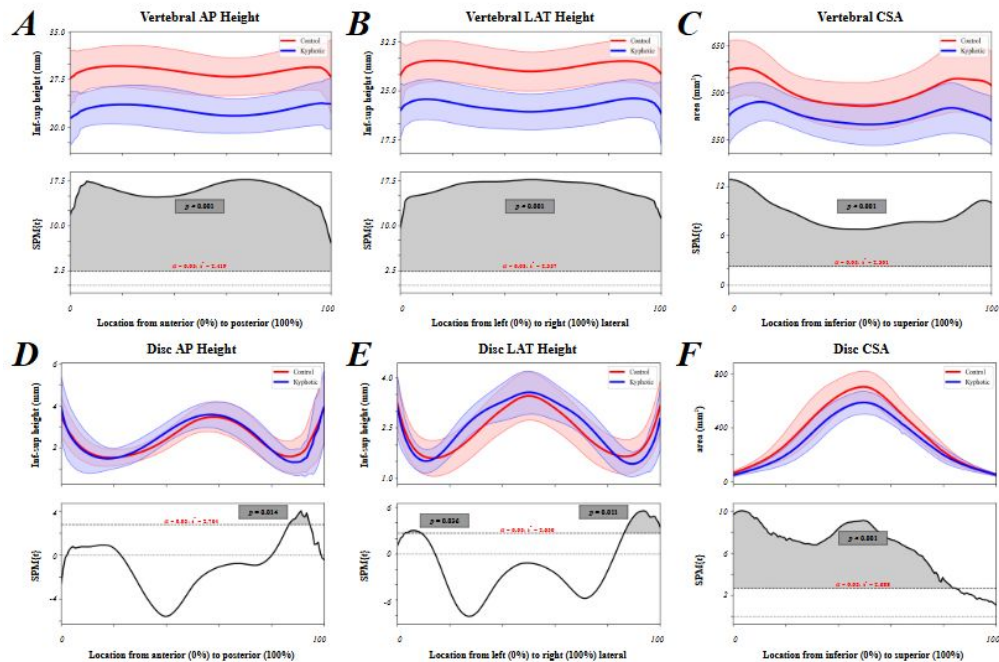
Spine-level scalar measurements



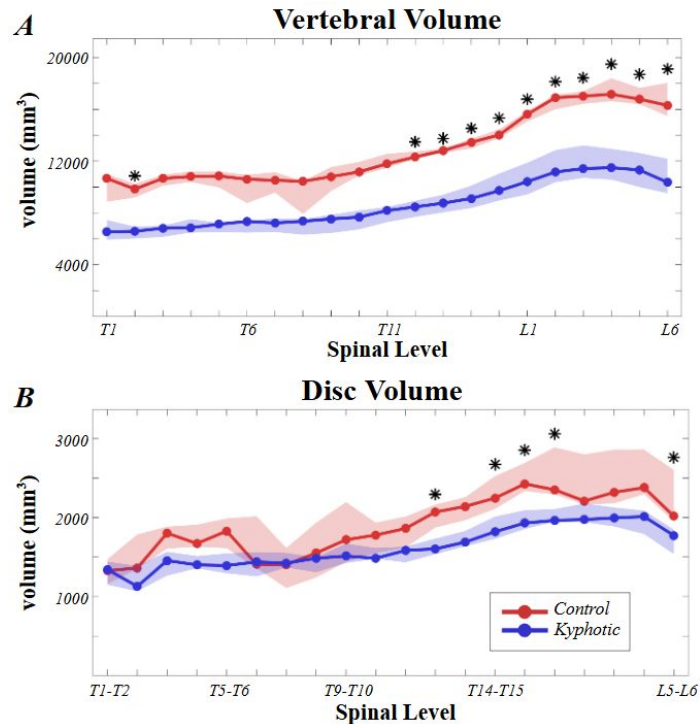
Summary of Results from *Morphologies*

Settings: # of slices = 200, height resolution = 200, slicer & height ignorance = 0.1, levels exported = T1 → L6

Spatially-varying measurements



Spine-level scalar measurements



***Morphologies* Future Improvements**

Points of Improvement:

1. Use a generalized naming convention for vertebra and disc levels to avoid the issue of variable total levels in porcine spines
2. Include more comprehensive and efficient visualization techniques
3. Separate the disc construction and morphology workflows
 - a. Moreover: include a workflow for construction of other sub-tissue structures like the NP and epiphyseal and growth endplates
4. Connect morphologies and geometry construction workflows to automated Gmsh meshing workflow to create full, node-conforming, watertight meshes for FEM